

POLS 3316: Statistics for Political Scientists

Homework 2: Chapters 4–6

Due Tuesday, October 13 at 11:59 PM · 100 points

Name: _____

Instructions

Questions 1–30 are multiple-choice questions. Points are shown for each question.

- For most questions, choose one best answer.
- For questions marked “Select ALL that apply” (Questions 5, 8, 18, 20, 23, 27, 29, and 30), more than one answer may be correct. Write all correct letters on the Answer Sheet. Partial credit is given for each correct letter.
- You may use the textbook, your notes, and lecture slides while completing this homework.
- Record your final answers on the Answer Sheet on the last page.

Calculation Questions: For Questions 25 and 28, use the space provided to show your work. Because this homework is multiple choice, you will not lose points if you do not show your work. However, showing your work is strongly recommended as practice for the exam. On the exam, you must show your work for calculation questions to receive full credit. If your work is missing or incomplete, you may receive only partial credit even if your final answer is correct.

Submission

You may complete the homework in either of the following ways:

1. **Print the homework and complete it by hand.** Then scan or take clear photos of all completed pages.
2. **Complete the homework electronically** by writing or marking your answers directly on the PDF.

When you are finished, **upload your completed homework to Canvas by the deadline.**

Make sure your **name is included** and that all answers are clearly visible before submitting.

Chapter 4: Research Design

1. [3 points] According to the textbook's definition, which of the following studies is an experiment?

- A. A researcher surveys a random sample of 1,000 voters and uses statistics to compare those who saw an ad with those who did not.
- B. A researcher invites students to a lab and lets each student choose whether to watch a campaign ad or a nature video.
- C. A researcher decides that half of the participants will watch a campaign ad and flips a coin to decide which half.
- D. A researcher analyzes a large dataset of election results from all 50 states with advanced statistical software.

2. [3 points] Why does an experiment need a control group?

- A. To make the sample representative of the whole population
- B. To let participants choose the treatment they prefer
- C. To measure Y more reliably than the treatment group can
- D. To show what would probably have happened without the treatment

3. [3 points] Chapter 4, Exercise 1 asks whether a researcher could realistically test a relationship with an experiment. For which relationship could a researcher realistically randomly assign X?

- A. An individual's religiosity (X) and his or her candidate preferences (Y)
- B. Military service (X) and attitudes toward foreign policy (Y)
- C. A speaker's personal characteristics, such as accent or dress (X), and persuasiveness (Y)
- D. A country's regime type (X) and its economic growth (Y)

4. [3 points] When we say that an experiment "isolates" the effect of X, what do we mean?

- A. Other causes of Y are balanced across groups, so the groups differ mainly in X.
- B. Every other cause of Y has been removed from the study.
- C. Only people with no other causes of Y are allowed to participate.
- D. The researcher measures Y only once, after the treatment is given.

5. [3 points] Which of the following statements about internal and external validity are accurate? Select ALL that apply.

- A. A study may have high internal validity even when its participants are not representative of the population researchers ultimately care about.
- B. Random sampling primarily strengthens internal validity by making treatment and control groups comparable.
- C. Random assignment can strengthen a causal claim within a study without establishing that the same effect would occur in other populations or settings.
- D. If a study has strong internal validity, its results must apply to all populations and settings.

6. [3 points] Match each study to its design. (1) Researchers randomly assign which wording of a question respondents see in an online poll. (2) Researchers randomly assign which households receive door-to-door canvassing during a real election. (3) Researchers compare towns just across a state line after a new election law applies to only one side. (4) Participants come to a campus room and watch randomly assigned videos.

- A. 1 = lab; 2 = field; 3 = natural; 4 = survey
- B. 1 = survey; 2 = field; 3 = natural; 4 = lab
- C. 1 = survey; 2 = natural; 3 = field; 4 = lab
- D. 1 = field; 2 = survey; 3 = natural; 4 = lab

7. [3 points] A researcher draws a random sample of 1,500 Texas voters. She asks each person (a) whether they saw a candidate's TV ad and (b) whether they plan to vote for the candidate. People who saw the ad are 8 points more supportive. She writes: "Because this is a random sample of Texas voters, the 8-point gap is the causal effect of the ad for all Texas voters." Which evaluation is best?

- A. Correct, because random sampling makes ad exposure unrelated to other causes of support.
- B. Correct only if each group, ad-viewers and non-viewers, contains at least 500 voters.
- C. Random sampling may let the 8-point gap generalize, but it does not make the gap causal.
- D. Incorrect, because a sample of 1,500 people can never represent all voters in Texas.

8. [4 points] A survey finds that people who saw a candidate's TV ad are more likely to say they will vote for the candidate (see Figure 4.2 from the textbook below). A researcher then runs an experiment in which a computer randomly decides who sees the ad. Which statements are correct? Select ALL that apply.

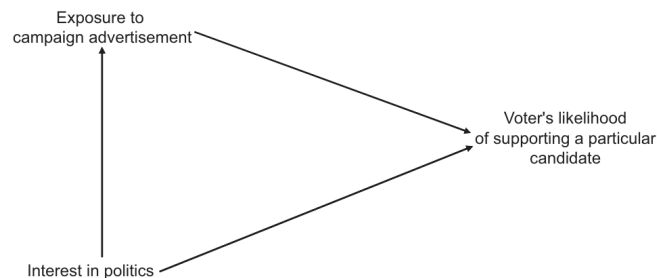


Figure 4.2 The possibly confounding effects of political interest in the advertisement viewing–vote intention relationship (K&W, p. 83)

- A. In the survey, political interest (Z) may affect both who sees the ad and vote intention.
- B. In the survey, people who already planned to vote for the candidate may have sought out the ad, so causal direction (Hurdle 2) is unclear.
- C. In the experiment, random assignment stops political interest from affecting vote intention.
- D. In the experiment, random assignment makes political interest unrelated to who sees the ad.

9. [4 points] A researcher compares the political priorities of citizens who follow political news every day with those who rarely do. To deal with selection bias, she controls for age, education, income, and party. Which statement is most accurate?
- A. Selection bias means her sample does not represent the population, so a larger random sample would solve the problem.
 - B. Once the four controls are included, selection bias is fully solved, because the two groups are now the same on every Z.
 - C. Selection bias cannot arise here, because citizens chose their own news habits rather than being assigned by a researcher.
 - D. The controls help only for the Zs she measured; unmeasured differences, such as motivation or social networks, may still affect Y.
10. [4 points] In an online survey experiment, respondents are randomly assigned to read either a news story about rising crime or a neutral story about gardening. The crime-story group reports more concern about crime. Which statement is most accurate?
- A. The difference can support a causal claim for these respondents because the story was randomly assigned.
 - B. The difference cannot be causal because the study was conducted online.
 - C. This is an observational study because respondents answered a survey.
 - D. Random assignment means the result automatically applies to all U.S. adults.

Chapter 5: Measuring Concepts of Interest

11. [3 points] Two research assistants independently code the same 200 news articles as favorable, neutral, or unfavorable toward a candidate, using the same coding rules. They agree on only 54 percent of the articles. This finding most directly raises a concern about:
- A. construct validity.
 - B. random sampling.
 - C. reliability.
 - D. external validity.
12. [3 points] According to Chapter 5, a measure is valid when it:
- A. gives the same result every time the same case is measured.
 - B. accurately represents the concept it is supposed to measure.
 - C. comes from a random sample of the population.
 - D. produces values that are never too high or too low.
13. [3 points] Kellstedt and Whitten compare two measures of the same variable. Measure A is biased but reliable; Measure B is unbiased but unreliable. For testing a theory about the relationship between X and Y, which do they prefer?
- A. Measure A, because a consistent distortion still lets the X–Y pattern show.
 - B. Measure B, because being correct on average matters most for theory testing.
 - C. Neither, because the two measures cause exactly the same problem.
 - D. Measure B, because random errors cancel out and make patterns clearer.

14. [3 points] To measure family economic resources, a survey asks college students, “How much money did you personally earn last year?” When the same students are asked again a week later, almost everyone gives the same answer. Which statement is most accurate?

- A. Because the answers are stable, the measure must be a valid measure of family resources.
- B. The measure is unreliable, because wealthy students who do not work report low earnings.
- C. The measure is biased, and a biased measure cannot also produce reliable answers.
- D. The measure appears reliable but may still be invalid for family economic resources.

15. [3 points] Pre-transfer income is income before government benefits such as food assistance and tax credits; post-transfer income is income after those benefits. Researcher 1 asks how well government programs reduce poverty. Researcher 2 asks how being poor shapes a person’s political attitudes. Which pairing of measures best fits each question?

- A. Researcher 1: pre-transfer income; Researcher 2: post-transfer income
- B. Researcher 1: post-transfer income; Researcher 2: pre-transfer income
- C. Both should use post-transfer income, because it is the only valid measure of poverty
- D. Both should use pre-transfer income, because transfers create measurement bias

16. [3 points] An early survey (Stouffer 1955) measured political tolerance by asking every respondent whether communists should be allowed to teach in public schools. Sullivan, Piereson, and Marcus (1979) argued that this was not a valid measure of tolerance. What was their main point?

- A. The question used too few groups, so the measure was unreliable.
- B. Tolerance means allowing rights to a group you oppose, but the question assumed every respondent opposed communists.
- C. Surveys cannot measure attitudes, so tolerance should be measured with experiments instead.
- D. The question should have been asked twice to check whether answers stayed the same.

17. [3 points] Consider two proposed measures (K&W Ch. 5, Exercise 3). Measure 1: political corruption in a country = the percentage of its politicians convicted of corrupt practices that year. Measure 2: democracy = whether the country’s constitution calls the nation “democratic.” Which evaluation is best?

- A. Measure 1 likely undercounts corruption, because many corrupt politicians are never convicted; Measure 2 is invalid, because nondemocracies can also call themselves “democratic.”
- B. Measure 1 is unreliable; Measure 2 is valid because a constitution is an official document.
- C. Both measures are valid because they come from official records.
- D. Both measures are unreliable because their values can change from year to year.

18. [4 points] A researcher finds that U.S. states with a larger share of college graduates also have a larger share of Democratic voters. Which statements are accurate? Select ALL that apply.

- A. The unit of analysis in this comparison is the state, not the individual voter.
- B. The state-level pattern shows that individual college graduates are more likely to vote Democratic.
- C. Random measurement error in the state education measure could make the relationship harder to detect.
- D. A clear conceptual definition of “state education level” should specify what characteristic of states is being measured.

19. [4 points] A new democracy index averages three components: (a) competitive elections, (b) civil liberties, and (c) adult literacy. The researcher finds that the index is strongly related to GDP per capita and claims this as evidence of construct validity. Suppose literacy itself is also strongly related to GDP per capita. What is the main concern?

- A. The relationship with GDP may partly reflect the literacy component already built into the democracy index.
- B. Construct validity requires a measure to produce exactly the same value every time.
- C. A measure cannot have construct validity if it combines several components.
- D. Any predicted relationship automatically proves that both the theory and the measure are correct.

20. [4 points] A researcher uses a computer program to score news articles as pro- or anti-candidate by counting positive and negative words from a fixed dictionary. Which statements are accurate? Select ALL that apply.

- A. If the same program gives the same scores every time it analyzes the same articles, the measure is highly reliable.
- B. If the measure is highly reliable, it must also be valid.
- C. If the program treats the phrase “a great failure” as positive because of the word great, the measure may be reliable but still invalid.
- D. If newly published articles receive different scores from older articles, that by itself shows the measure is unreliable.

Chapter 6: Getting to Know Your Data

21. [3 points] Figure 6.2 from the textbook (below) shows respondents' religious identification in the 2004 NES. Because religious identification is a categorical variable, which measure of central tendency is appropriate?

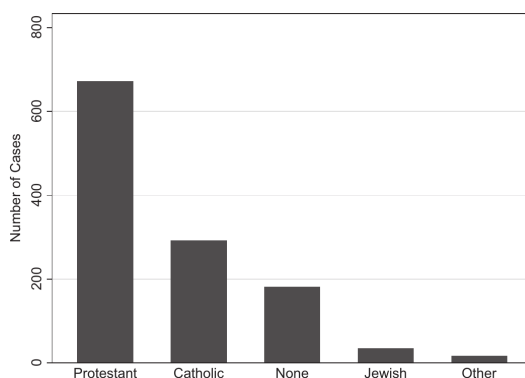


Figure 6.2 Bar graph of religious identification, NES 2004 (K&W, p. 132)

- A. The mean of the category codes
 - B. The mode: Protestant
 - C. The median: Catholic
 - D. The standard deviation
22. [3 points] A histogram of individual campaign donations shows that most donors gave less than \$100, but a few donors gave several thousand dollars. Which statistic best describes the typical donation?
- A. The mean, because it uses every observation.
 - B. The mode, because dollar amounts are categorical.
 - C. The standard deviation, because it measures the center of the distribution.
 - D. The median, because a few very large donations can pull the mean upward.
23. [3 points] Which of the following variables are ordinal? Select ALL that apply.
- A. Religious identification: Protestant, Catholic, Jewish, None, Other
 - B. Highest level of education completed: less than high school, high school diploma, some college, bachelor's degree, graduate degree
 - C. Satisfaction with government: very satisfied, somewhat satisfied, somewhat dissatisfied, very dissatisfied
 - D. State of residence
24. [3 points] A survey measures family financial situation using the following categories: 1 = Much better, 2 = Somewhat better, 3 = Same, 4 = Somewhat worse, 5 = Much worse
- Why is this variable best described as ordinal rather than continuous?
- A. The categories have a meaningful order, but we cannot assume that the difference between each pair of adjacent categories is equal.
 - B. The categories cannot be ranked from better to worse.
 - C. The variable is continuous because the categories are numbered from 1 to 5.
 - D. The variable is categorical because all survey responses are categorical.

25. [3 points] The number of political meetings attended by eight respondents last month is:

0, 1, 1, 2, 2, 2, 4, 8

What are the mean and the mode of this variable?

- A. Mean = 2.0; Mode = 2
- B. Mean = 2.5; Mode = 2
- C. Mean = 2.5; Mode = 1
- D. Mean = 3.0; Mode = 2

Show your work:

26. [3 points] A dataset has nine observations. Because of a data-entry error, the largest value is recorded as 450 instead of 45. Which statistics are likely to change the most?

- A. The median and IQR
- B. The mean and standard deviation
- C. The median only
- D. The mean, median, IQR, and standard deviation by about the same amount

27. [4 points] Which statements about measurement metrics are correct? Select ALL that apply.

- A. Party identification coded Republican / Independent / Democrat can be treated as ordinal only if Independents meaningfully lie between Republicans and Democrats.
- B. Numbering religious categories from 1 to 5 automatically makes religious identification ordinal.
- C. Treating an ordinal variable as continuous assumes equal unit differences.
- D. A variable can be continuous even if it does not have a meaningful zero.

28. [4 points] Five cities have the following values:

2, 4, 6, 8, 10

Using the Chapter 6 formula for sample variance,

$$s^2 = \frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{n - 1},$$

what is the variance?

- A. 8
- B. 10
- C. 16
- D. 40

Show your work:

29. [4 points] Figure 6.6 from the textbook (below) shows two histograms of the same variable, incumbent-party presidential vote percentage, 1876–2016. The top histogram uses 2 bins, and the bottom histogram uses 10 bins. The two graphs look noticeably different. Which statements are correct? Select ALL that apply.

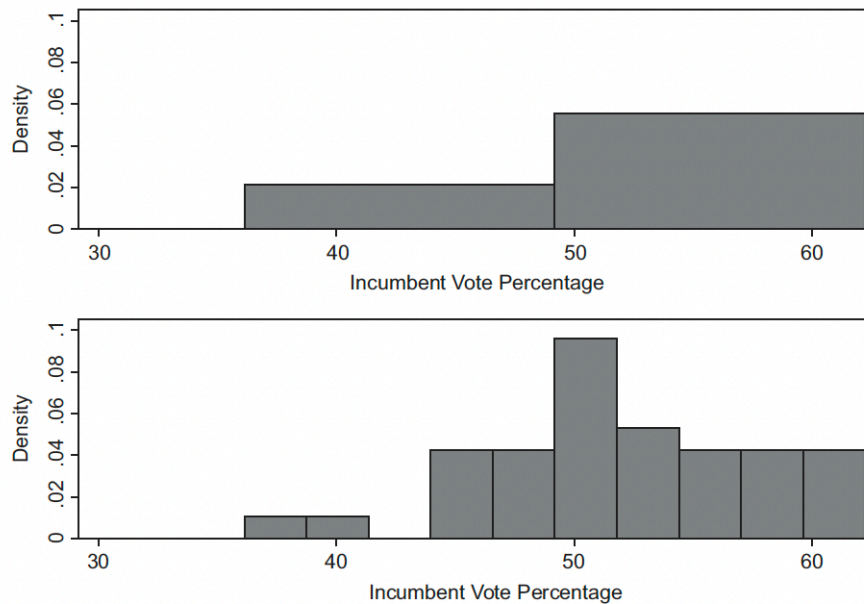


Figure 6.6 Histograms of incumbent-party presidential vote percentage, 1876–2016, depicted with two and then ten blocks (K&W, p. 138)

A. Both histograms can be correct even though they look different.
 B. Changing the number of bins changes the underlying observations in the dataset.
 C. The number of bins can affect how the shape of the distribution appears.
 D. If two histograms from the same data look different, one of them must be incorrect.

30. [4 points] A continuous variable has a 25th percentile of 20 and a 75th percentile of 32. Which statements are correct? Select ALL that apply.

- A. The interquartile range (IQR) is 12.
- B. The median must be 26.
- C. The middle 50 percent of the observations fall between 20 and 32.
- D. The IQR is a measure of central tendency.

Answer Sheet

Name: _____

Write the letter of your answer for each question.

For “Select ALL” questions, write every correct letter.

- | | | | | |
|-----------|-----------|-----------|-----------|-----------|
| 1. _____ | 2. _____ | 3. _____ | 4. _____ | 5. _____ |
| 6. _____ | 7. _____ | 8. _____ | 9. _____ | 10. _____ |
| 11. _____ | 12. _____ | 13. _____ | 14. _____ | 15. _____ |
| 16. _____ | 17. _____ | 18. _____ | 19. _____ | 20. _____ |
| 21. _____ | 22. _____ | 23. _____ | 24. _____ | 25. _____ |
| 26. _____ | 27. _____ | 28. _____ | 29. _____ | 30. _____ |